

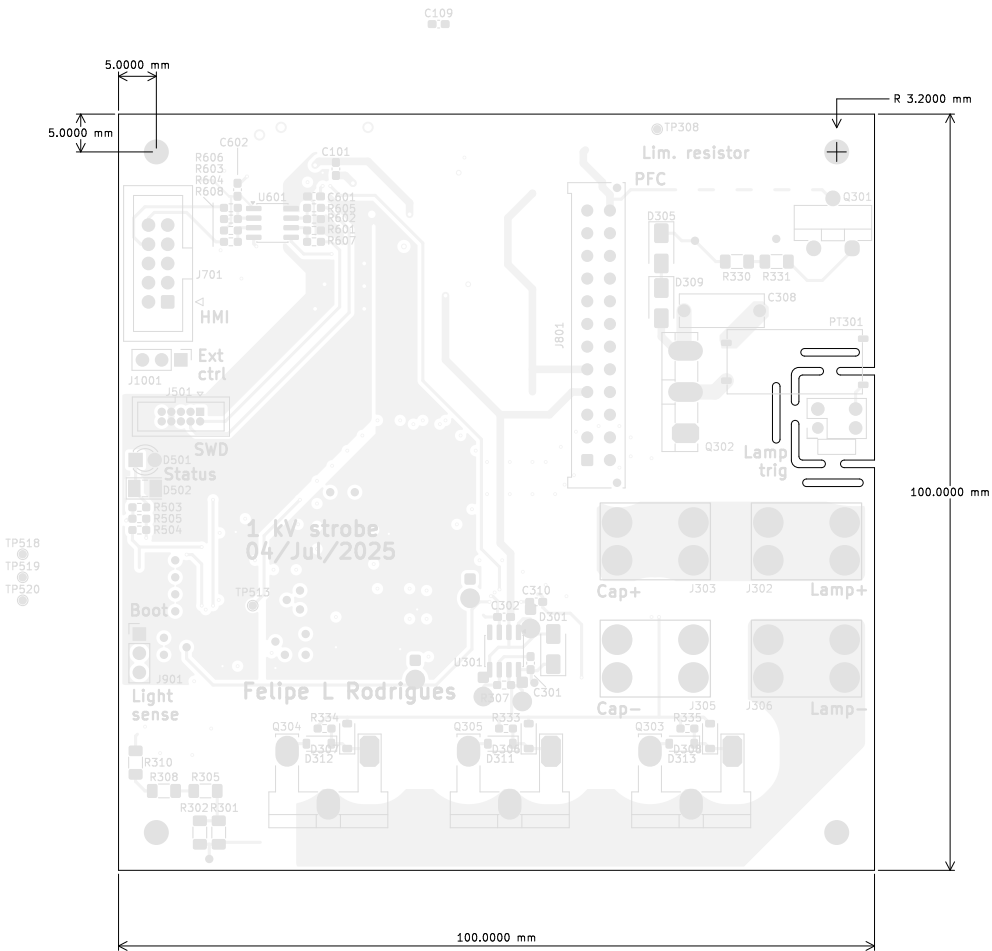
Main board Fabrication Document

Top Fabrication (Scale 1:1)

Layer Stack Legend

Material	Layer	Thickness	Dielectric	Type	Gerber
F.Paste				Paste Mask	
F.Silkscreen				Legend	GBR
F.Mask		0.02mm	Solder Resist	Solder Mask	GBR
Copper	F.Cu (Sig, PWR)	0.07mm (2.00oz)		Signal	GBR
Prepreg		0.1mm	FR4_7628	Dielectric	
Copper	In1.Cu (GND)	0.035mm (1.00oz)		Plane	GBR
Core		1.828mm	FR4	Dielectric	
Copper	In2.Cu (Sig, PWR)	0.035mm (1.00oz)		Plane	GBR
Prepreg		0.1mm	FR4_7628	Dielectric	
Copper	B.Cu (Sig, PWR)	0.07mm (2.00oz)		Signal	GBR
B.Mask		0.02mm	Solder Resist	Solder Mask	GBR
B.Silkscreen			Direct Printing	Legend	GBR
B.Paste				Paste Mask	

Total thickness: 2.278mm
Note: external layer thicknesses are specified after plating



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts,GBR" SUFFIX.
DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.
SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 4) SURFACE FINISH: HAL SNP8
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR GREEN.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:
 - A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
 - B. Tg 170 C OR EQUIVALENT.
 - C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY CHURRASCOD ELECTRONICS.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

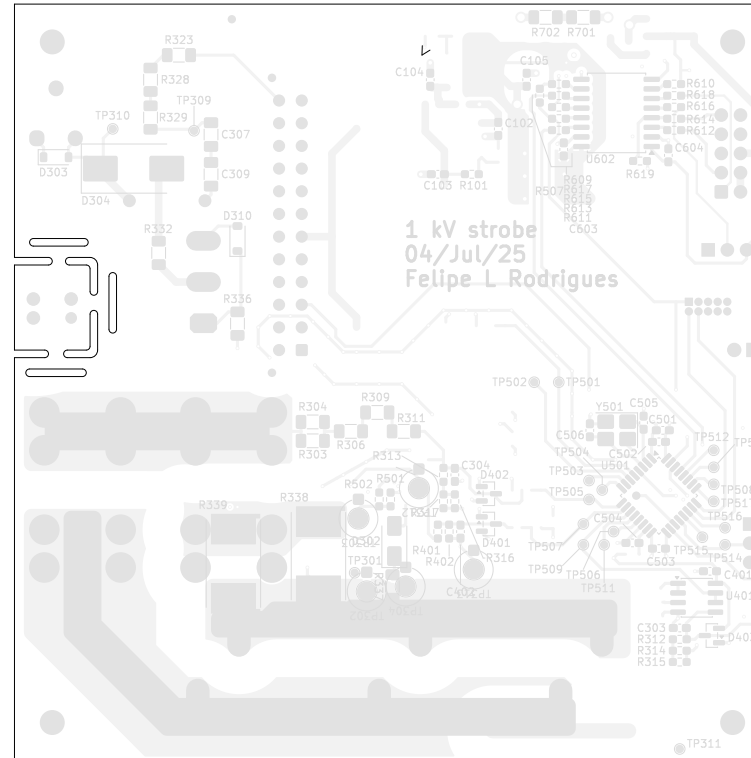
BOARD SIZE	100.000 x 100.000 mm
BOARD THICKNESS	2.278 mm
TRACE WIDTH	0.300 mm
TRACE TO TRACE	0.200 mm
MIN. HOLE (PTH)	0.300 mm
MIN. HOLE (NPTH)	1.000 mm
ANNULAR RING	0.100 mm
COPPER TO HOLE	0.300 mm
COPPER TO EDGE	0.500 mm
HOLE TO HOLE	0.250 mm



Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 9773310
	Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Title: Top Fabrication (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13
Sheet Path:	Reviewer:		Size: A3	Revision: 1.0.0+ (Unreleased)
			Sheet: 1 of 10	

Main board Fabrication Document

Bottom Fabrication (Scale 1:1)



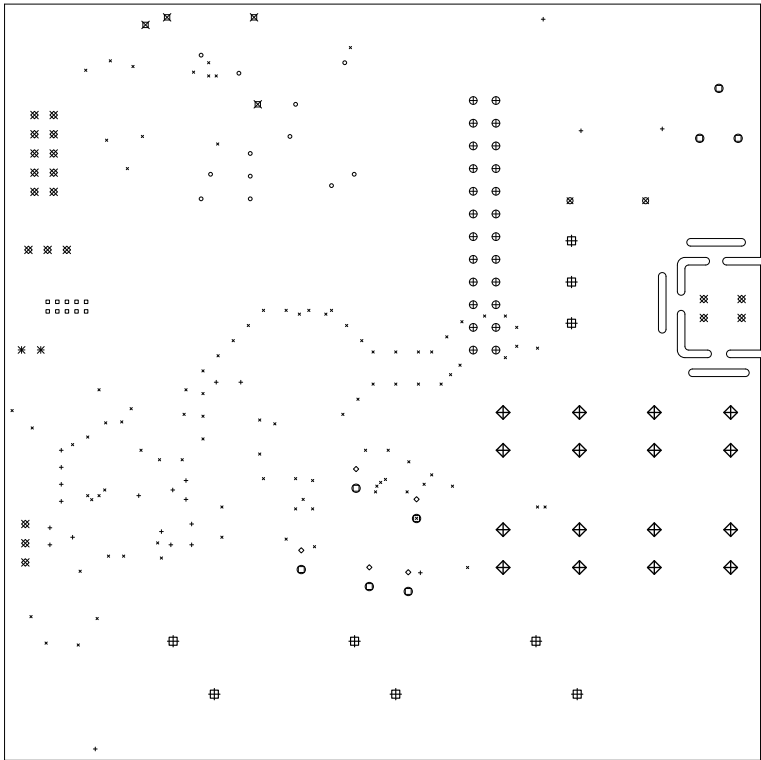
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	Board Name: Main board		Project Name: 1 kV strobe	
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 2 of 10

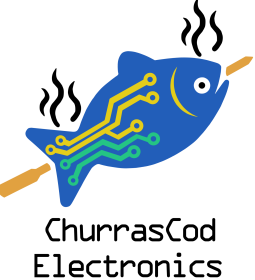
Main board Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	101	0,30mm (11,81mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Via
○	12	0,50mm (19,69mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Via
+	25	0,50mm (19,69mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
□	10	0,65mm (25,59mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
◇	9	0,70mm (27,56mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊗	6	0,80mm (31,50mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
※	2	0,90mm (35,43mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊠	4	1,00mm (39,37mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Via
⊗	20	1,00mm (39,37mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊕	24	1,02mm (40,16mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
○	8	1,10mm (43,31mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
○	2	1,20mm (47,24mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊕	9	1,50mm (59,06mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
⊕	16	1,80mm (70,87mils)	PTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Pad
Total 248						



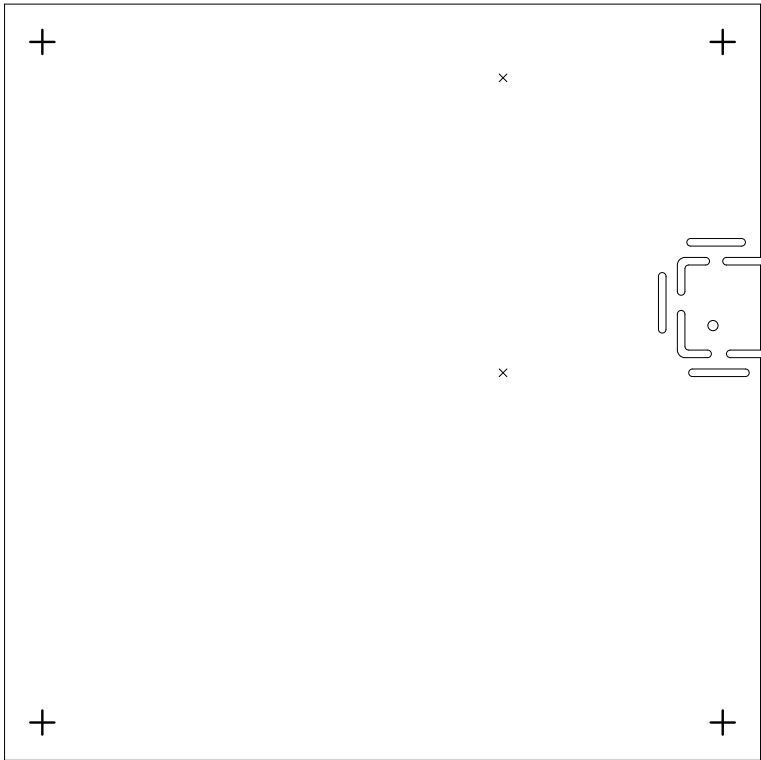
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			Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)	
	Sheet Path:		Reviewer:	Size: A3	Sheet: 3 of 10	

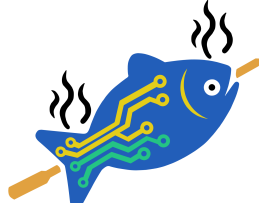
Main board Fabrication Document

Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

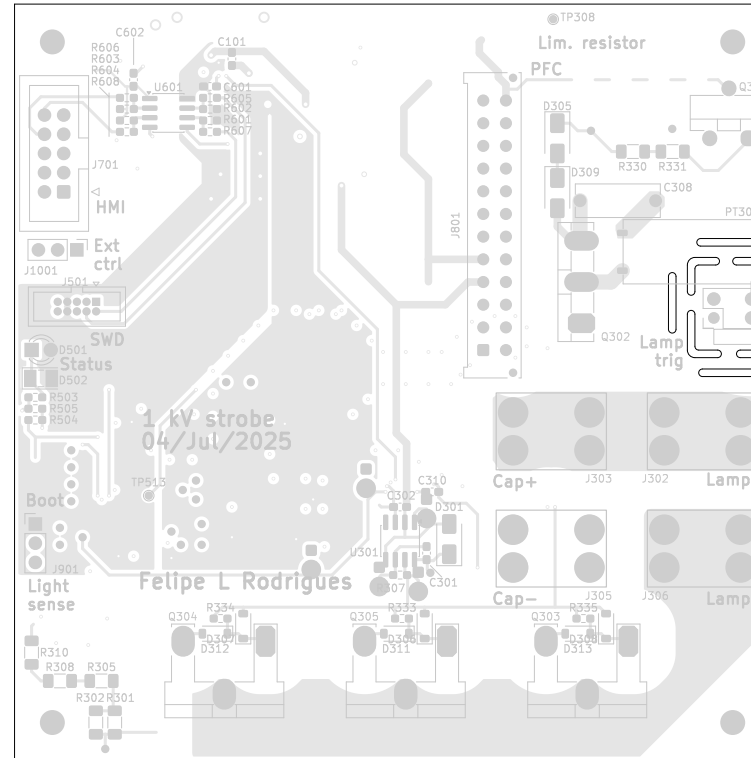
Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	2	1,02mm (40,16mils)	NPTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Mechanical
○	1	1,35mm (53,15mils)	NPTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Mechanical
+	4	3,20mm (125,98mils)	NPTH	Round	F,Cu (Sig, PWR) - B,Cu (Sig, PWR)	Mechanical
Total 7						



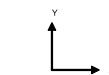
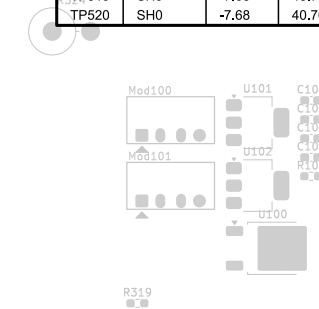
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			Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: Strobe_main.kicad_pcb		Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
	Sheet Path:		Reviewer:	Size: A3	Sheet: 4 of 10	

Main board Fabrication Document

Top Test Points (Scale 1:1)



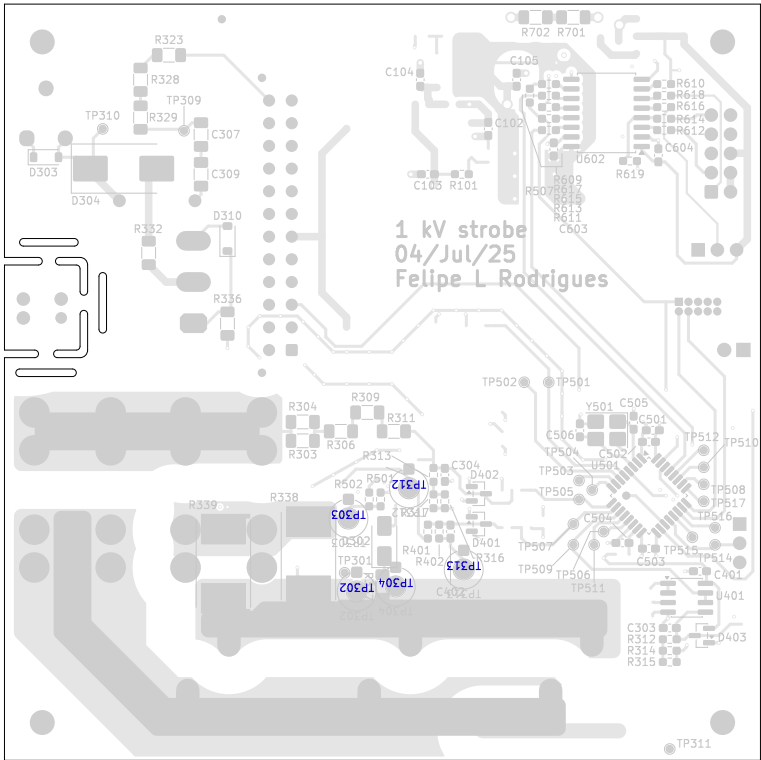
Ref.	Net	X [mm]	Y [mm]
TP308	S1k	76.25	103.00
TP513	Vbus_ctrl	22.75	40.00
TP518	BOOT0	-7.68	46.80
TP519	SH0	-7.68	43.75
TP520	SH0	-7.68	40.70



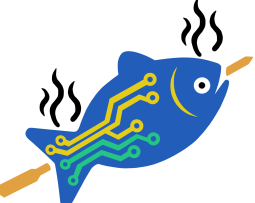
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	Board Name: Main board	Project Name: 1 kV strobe		
Sheet Title: Top Test Points (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 5 of 10

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Bottom Test Points (Scale 1:1)

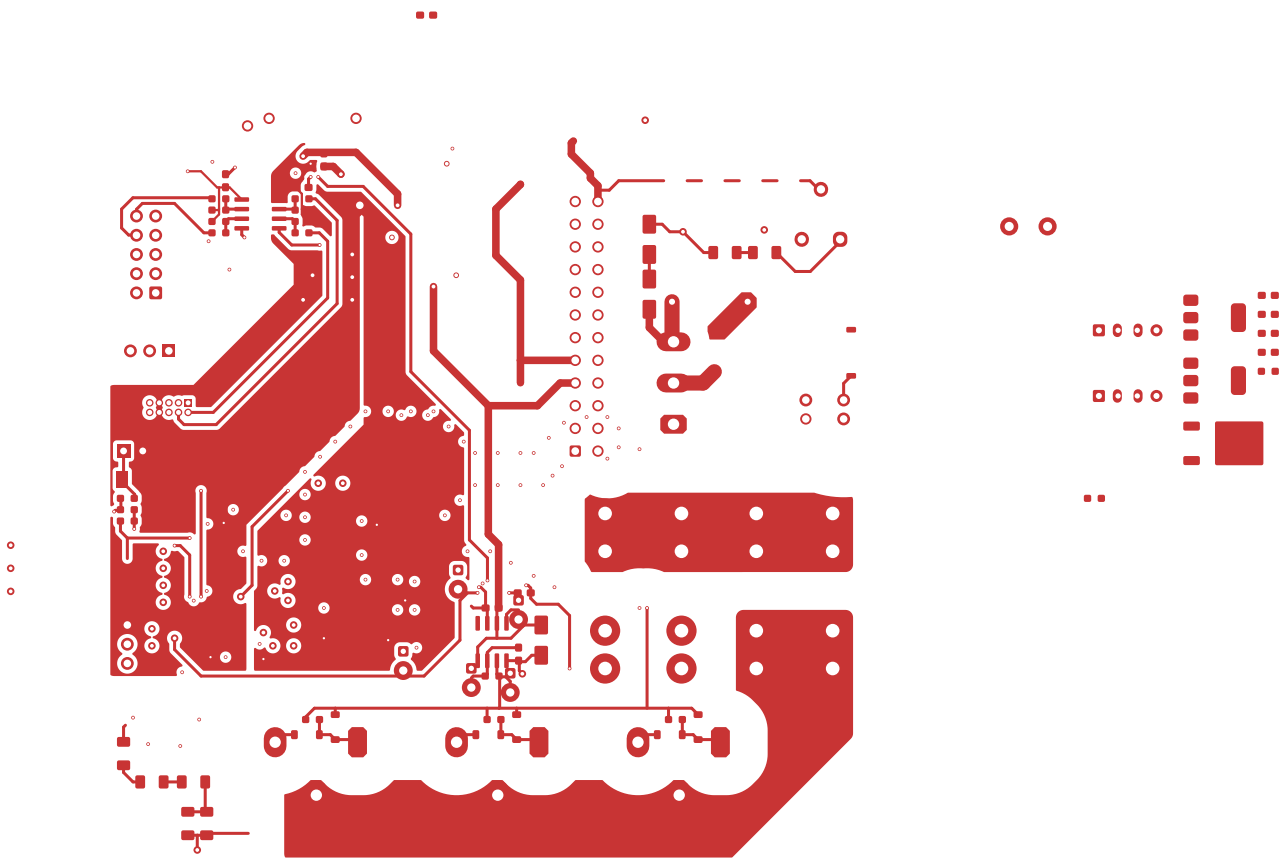


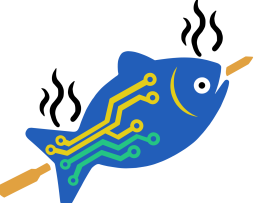
Ref.	Net	X [mm]	Y [mm]
TP301	VBS	60.00	29.80
TP302	LMP_EN	58.40	28.60
TP303	SH0	59.50	38.25
TP304	Mon_S1k	53.25	29.25
TP309	TP_REF	81.25	88.25
TP310	TP_PRI	92.00	88.50
TP311	LMP	17.00	6.50
TP312	SH0	51.50	42.25
TP313	DRV_OUT	44.25	31.50
TP501	Net-(U501-PA0)	33.00	55.00
TP502	Net-(U501-PA1)	36.25	55.00
TP503	SUP_TX	29.00	42.00
TP504	SUP_RX	27.25	40.75
TP505	Mon_S1k_buf	29.00	39.50
TP506	Net-(U501-PB0)	25.75	35.25
TP507	Mon_IL_prot	29.75	36.25
TP508	Net-(U501-PB3)	12.50	41.50
TP509	Mon_LMP_buf	29.75	33.50
TP510	Net-(U501-PB4)	12.50	43.75
TP511	Net-(U501-PA7)	27.00	33.50
TP512	Net-(U501-PB5)	12.50	46.00
TP514	Light_sns	11.00	33.50
TP515	INT_EN	14.00	34.50
TP516	Net-(U501-PA12)	11.00	35.75
TP517	Net-(U501-PA15)	12.50	39.25

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	Sheet Title: Bottom Test Points (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
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Main board Fabrication Document

F.Cu (Sig, PWR) (Scale 1:1)

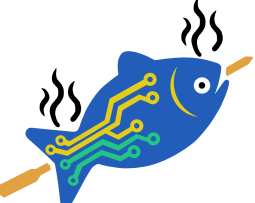


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		Board Name: Main board		Project Name: 1 kV strobe	
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Main board Fabrication Document

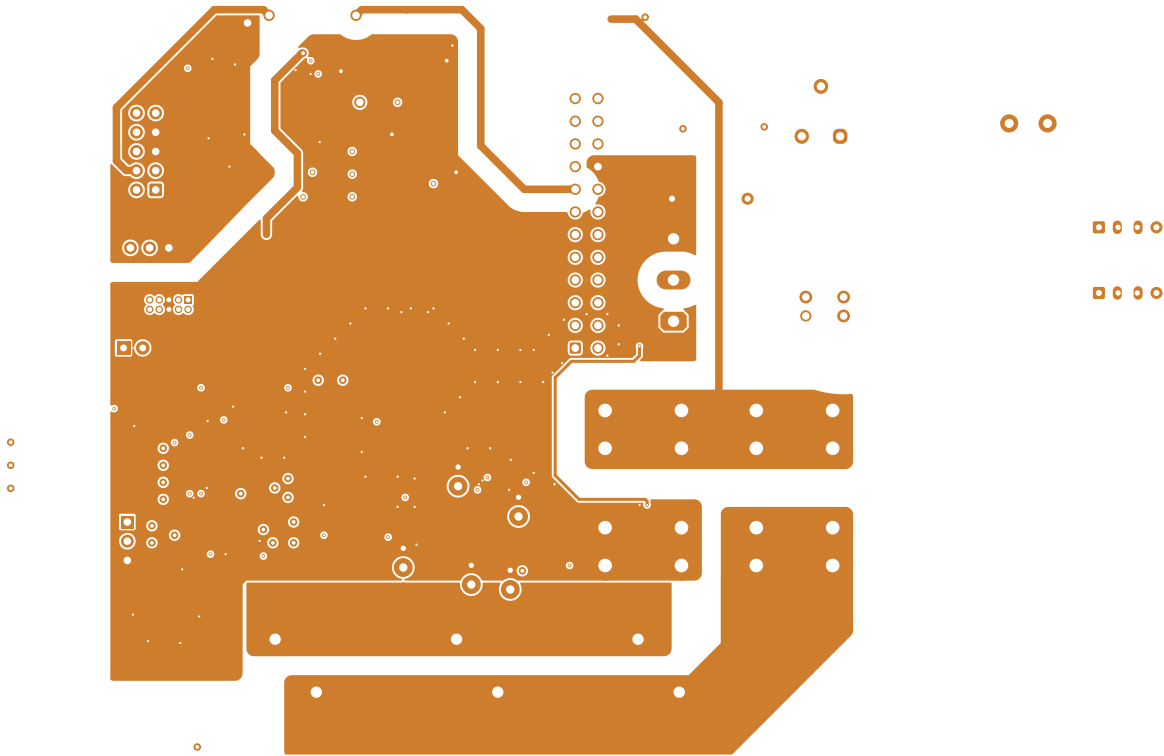
In1.Cu (GND) (Scale 1:1)

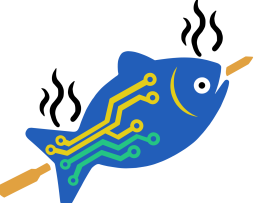


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	Sheet Title: In1.Cu (GND) (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
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Main board Fabrication Document

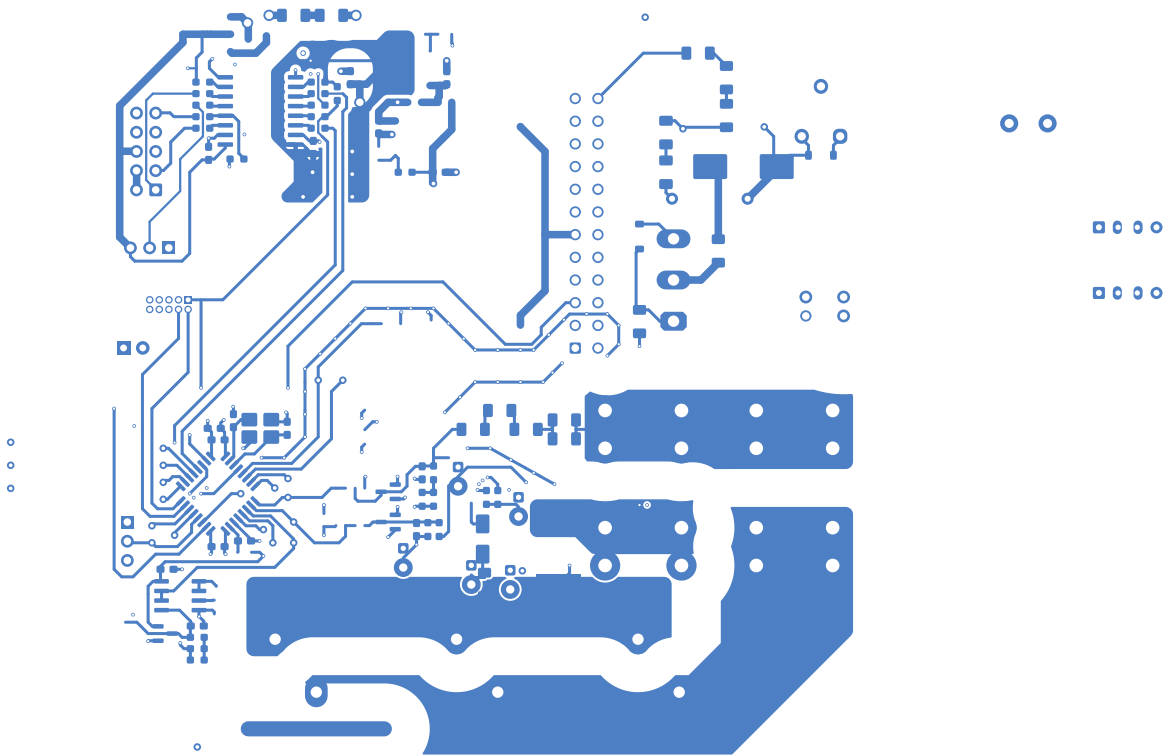
In2.Cu (Sig, PWR) (Scale 1:1)

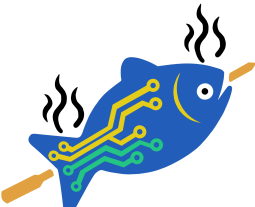


 ChurrasCod Electronics	Comments:	Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 9773310
	Sheet Title: In2.Cu (Sig, PWR) (Scale 1:1)	File Name: Strobe_main.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
	Sheet Path:	Reviewer:		Size: A3	Sheet: 9 of 10

Main board Fabrication Document

B.Cu (Sig, PWR) (Scale 1:1)



 ChurrasCod Electronics	Comments:		Company: ChurrasCod Electronics		Variant: PRELIMINARY	Git Hash: 9773310
			Board Name: Main board		Project Name: 1 kV strobe	
	Sheet Title: B.Cu (Sig, PWR) (Scale 1:1)	File Name: Strobe_main.kicad_pcb		Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
	Sheet Path:		Reviewer:	Size: A3	Sheet: 10 of 10	